

ST-CoSense: Spatio-Temporal Cooperative Sensing for Resource-Constrained Urban Applications

Yaxin Mei, Huiling Qin, Guangxue Zhang, Xiaolong Xu, *Senior Member, IEEE*, Zhanchuan Cai, *Senior Member, IEEE*, Yu Zheng, *Fellow, IEEE*, and Tian Wang, *Senior Member, IEEE*

Abstract—Urban sensing is a critical infrastructure for smart cities, yet dense sensor deployments remain prohibitively costly, and sensor data quality degrades severely under real-world interference (e.g., signal occlusion, extreme weather) that results in data gaps. Existing approaches treat sampling and imputation as two independent stages: sampling strategies select locations based on heuristics or optimization criteria without considering the downstream reconstruction objective, while imputation models passively adapt to whatever observations are provided, with no mechanism to inform the sampler which locations would most effectively reduce reconstruction error. We propose ST-CoSense, a unified framework that bridges this gap through end-to-end joint optimization of active sparse sampling and spatio-temporal imputation. A mask-guided encoder dynamically selects informative sensing locations by integrating historical sampling patterns with learned spatio-temporal representations, while an attention-based imputation module reconstructs complete urban data from sparse observations. Imputed results in turn refine subsequent sampling decisions, allowing continuous adaptation under sampling budget constraints. Experimental results show that ST-CoSense outperforms state-of-the-art methods, achieving up to 9.07% MAE and 10.53% RMSE improvement at a 50% sampling rate. Furthermore, ST-CoSense demonstrates superior robustness in large-scale urban networks, achieving a 4.95% RMSE improvement over the best-performing baseline. These results underscore ST-CoSense’s effectiveness and its practical potential in resource-constrained urban sensing applications.

Index Terms—Urban Sensing, Active Sparse Sampling, Spatio-Temporal Imputation, End-to-End Co-Optimization.

I. INTRODUCTION

Urban sensing systems [1] serve as a critical infrastructure component for smart cities, enabling transformative applica-

tions across transportation management, environmental governance, and public service optimization. Multi-dimensional urban sensing streams (e.g., vehicular mobility patterns and atmospheric indices) have been shown to substantially improve urban governance through three-stage data processing pipelines: sensing, analytics, and execution [2]. As demonstrated in [3], [4], adaptive signal control systems leveraging real-time Vehicle-To-Infrastructure (V2I) data can achieve congestion reduction across multiple city areas. Similarly, works [5], [6] realize early warning of air pollution and location of pollution sources through air quality sensing, which can effectively assist the formulation of emission strategies. Such applications exemplify urban sensing’s potential to reshape city governance through data-driven feedback loops.

The traditional pattern of urban sensing systems predominantly relies on dense sensor deployments to ensure comprehensive data coverage, commonly measured by the coverage ratio [7], which serves as the fundamental prerequisite for subsequent analytical processes and Quality of Service (QoS) assurance. This pattern typically incorporates statistical, machine learning, and deep learning methods to compensate for inevitable data gaps caused by various factors, including signal interference in complex urban environments and sensor performance degradation during extreme weather conditions [8]. However, this “sense-then-impute” paradigm presents two critical issues. First, achieving comprehensive spatial coverage through dense sensor deployments incurs prohibitive costs in city-scale implementations, including substantial capital investment for hardware procurement and installation, along with ongoing operational expenses for maintenance, calibration, and data transmission [9]. Second, and more fundamentally, the decoupled design of sensing and imputation processes limits overall system performance. Sampling strategies in sensing are designed using heuristics or optimization criteria (e.g., uncertainty-based selection and entropy maximization) that are independent of the imputation model’s reconstruction objective. This separation means that locally informative locations are not necessarily globally optimal for reconstruction. For instance, a policy favoring high-variance locations may oversample volatile hotspots while neglecting spatially diverse observations that better constrain the imputation through inter-location correlations. Existing works [10] have demonstrated that integrating sampling and imputation with iterative feedback significantly outperforms static approaches, establishing that joint optimization yields substantially better sensing performance.

To solve the above two issues, a more feasible and cost-

Yaxin Mei is with Institute of Artificial Intelligence and Future Networks, Beijing Normal University, Zhuhai 519087, China; School of Artificial Intelligence, Beijing Normal University, Beijing 100875, China. (E-mail: meiyax@163.com)

Huiling Qin and Guangxue Zhang are with Institute of Artificial Intelligence and Future Networks, Beijing Normal University, Zhuhai 519087, China. (E-mail: orekinana@gmail.com; guangxue_zhang@bnu.edu.cn)

Xiaolong Xu is with the School of Software, Nanjing University of Information Science and Technology, Nanjing 210044, China. (E-mail: xlxu@ieee.org)

Zhanchuan Cai is with School of Computer Science and Engineering, Macau University of Science and Technology, Macau, 999078, China. (E-mail: zccai@must.edu.mo)

Yu Zheng is with the School of Computing and Artificial Intelligence, Southwest Jiaotong University, Chengdu 611756, China; School of Cyber Engineering, Xidian University, Xi’an 710126, China; Beijing Key Laboratory of Traffic Data Mining and Embodied Intelligence, Beijing 100044, China. (E-mail: msyuzheng@outlook.com)

Tian Wang is with Engineering Research Center of Cloud-Edge Intelligent Collaboration on Big Data, Ministry of Education, Beijing Normal University, Zhuhai 519087, China; College of Computer and Data Science, Fuzhou University, China; Computer Science and Mathematics, Fujian University of Technology, 350118, Fuzhou, China. (E-mail: cs_tianwang@163.com)

(Corresponding Author: Tian Wang.)

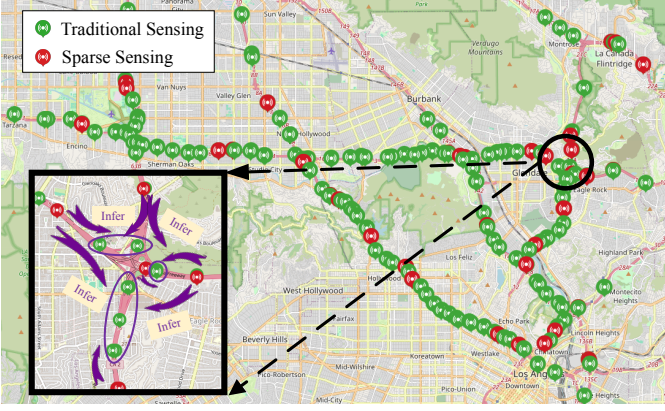


Fig. 1: Comparison of Traditional and Sparse Sensing on the METR-LA (Metro Traffic Los Angeles) Dataset [11]. Green markers indicate the sensor layout adopted by traditional sensing schemes, which rely on exhaustive spatial coverage to guarantee data completeness. Red markers indicate the sensor layout adopted in the sparse sensing paradigm, which strategically deploys a few sensors. By exploiting the inherent spatio-temporal correlations in urban data, sparse sensing techniques infer data in unsensed subareas (imputation process), significantly reducing reliance on dense sensor networks while preserving high-quality reconstruction.

effective strategy is to adopt sparse sensing, which has garnered substantial attention in the field of sparse Mobile Crowd Sensing (MCS) [12]. As illustrated in Fig. 1, this method involves a strategic deployment of a few sensors based on data characteristics and application-specific requirements while fully leveraging urban data’s inherent spatio-temporal continuity to enable accurate information reconstruction through advanced imputation technologies [13]. Despite its theoretical advantages in resource efficiency, implementing this paradigm in real-world scenarios faces three fundamental challenges:

- **How to determine an optimal set of sampling locations?**

The selection of sampling locations significantly influences the information gain for subsequent imputation processes, directly determining imputation accuracy. Without prior knowledge of future data distributions or evolving patterns, identifying optimal sampling strategies presents a fundamental challenge in sparse sensing.

- **How to achieve high-accuracy data imputation under sparse sampling?**

Sparse sampling inherently reduces the volume of observed data, posing significant challenges for models to accurately reconstruct complex spatio-temporal dependencies. The core challenge lies in developing robust models that can extract key patterns from limited observations while effectively bridging substantial data gaps, ensuring high-accuracy imputation.

- **How to integrate sparse sampling and data imputation into a unified framework?**

Jointly optimizing sampling and imputation is difficult due to the need for dynamic adaptation between the

two processes. Sampling must align with the requirements of imputation to maximize reconstruction accuracy. Meanwhile, the imputation model must fully leverage the sparse sampling data while providing effective feedback to inform the sampling strategy. Coordinating the interactions between sampling and imputation to form an end-to-end system is the key to this problem.

To address these challenges, we propose a unified framework ST-CoSense (Spatio-Temporal Cooperative Sensing) that integrates two core modules: active sparse sampling and spatio-temporal imputation. Active sparse sampling refers to the adaptive selection of a limited subset of sensing locations. This approach aims to maximize information gain for subsequent imputation under strict budget constraints. During the active sparse sampling phase, a spatio-temporal encoder is employed to extract latent spatio-temporal features of observed data. Building on this, the mask-guided encoder synthesizes a dynamic sampling matrix that adaptively integrates spatio-temporal patterns and historical sampling distributions, enabling context-aware optimization of sensor placement. In the spatio-temporal imputation stage, ST-CoSense employs spatio-temporal attention mechanisms to capture both local and global dependencies across spatial and temporal dimensions. These mechanisms dynamically weight the influence of neighboring subareas and temporal contexts, enabling the model to accurately infer the missing value even under low sampling rates. Through iterative refinement with multi-layer attention computations, ST-CoSense ensures robust imputation, which in turn enhances the representativeness of subsequent sampling iterations. This closed-loop design not only improves imputation accuracy but also establishes a synergistic relationship between sampling and imputation, making the framework highly adaptive to diverse urban sensing scenarios. Our key contributions are as follows:

- A unified framework ST-CoSense is proposed that jointly optimizes active sparse sampling and spatio-temporal imputation through closed-loop integration. Sampling locations are dynamically selected by the mask-guided encoder to minimize reconstruction error via end-to-end training, ensuring that sampling decisions directly enhance imputation quality.
- Spatio-temporal learners are designed to capture spatial correlations through learnable subarea embeddings and temporal dynamics through a dual-attention mechanism. The temporal learner recovers full temporal resolution from compressed representations and models long-range dependencies efficiently.
- ST-CoSense achieves superior performance across multiple urban spatio-temporal datasets (including the large-scale urban dataset), demonstrating robust sensing accuracy even under extremely low sampling rates.

II. RELATED WORK

This section reviews two lines of research that directly underpin ST-CoSense: sparse mobile crowd sensing and data imputation.

A. Sparse Mobile Crowd Sensing

Mobile crowd sensing uses smart devices carried by mobile users to perform various sensing tasks [14]. To achieve accurate and comprehensive sensing coverage, traditional MCS methods typically require recruiting a large number of participants in the target sensing area. However, due to constraints such as limited budgets, recruiting users to cover the whole target area is often impractical. To address these limitations, sparse MCS has recently emerged as a novel paradigm, strategically selecting only a subset of mobile users to perform sensing in key locations while employing imputation techniques to infer data for unsensed subareas [12].

Recently, many urban sensing systems have been developed using sparse MCS methods. Liu et al. [15] exploit spatial correlations of pollutant concentrations to formulate air quality monitoring as a compressive sensing problem, using active learning to select the most informative sensing locations and thus reduce recruitment costs while maintaining estimation quality. Similarly, Zhu et al. [16] identify low-rank structures in urban traffic data through PCA and use compressive sensing to reconstruct city-wide traffic conditions from sparse vehicle observations. These works highlight the practical potential of sparse MCS in achieving cost-effective urban sensing.

A central challenge in sparse MCS is determining which subareas to sense, as this decision directly governs the quality of downstream data recovery. Existing approaches address this challenge at different levels of coupling between sampling and imputation. Early works treat the two stages as largely decoupled. Xie et al. [7] propose an active sparse MCS framework based on matrix completion that selects informative locations by evaluating their contribution to the rank condition of the data matrix, and then performs imputation based on the resulting observations. Wang et al. [17] exploit spatio-temporal correlations within and between data to dynamically select a minimal set of subareas per cycle, using compressive sensing and active learning jointly to infer unobserved regions under a probabilistic quality guarantee. Zhu et al. [18] further relax the uniform-cost assumption by proposing a three-step cell selection method (information modeling, cost estimation, and joint cost-quality optimization) to select sensing locations under heterogeneous sensing costs. More recently, reinforcement learning has introduced an implicit coupling between sampling and imputation. Liu et al. [10] integrate matrix completion with reinforcement learning to prioritize critical sensing areas, where the reward signal reflects completion accuracy. Zeng et al. [19] propose a subarea evaluation method based on deep reinforcement learning that jointly considers the temporal effectiveness of sensing tasks and the informativeness of subareas for data inference. Although these methods allow imputation quality to indirectly influence sampling through reward shaping, the imputation model itself does not participate in the update of the policy gradient, and the two modules operate in separate representational spaces.

Across all these methods, the sampling criterion, whether derived from matrix completion theory, a learned reward signal, or coverage objectives, is defined without direct gradient-level reference to the downstream imputation model's recon-

struction objective. Once sensing locations are determined, the imputation stage operates on the resulting observations as a fixed input, with no pathway for reconstruction gradients to refine future sampling decisions.

In other domains, tighter coupling has proven beneficial. Zhang et al. [20] propose sequential active acquisition for MRI, where each sampling decision is conditioned on previous observations and intermediate reconstructions. Bahadir et al. [21] parameterize the MRI under-sampling pattern as a learnable mask and co-optimize it with reconstruction end-to-end. Liu et al. [22] co-optimize sensor placement and neural field recovery through bilevel differentiable learning. Inspired by these findings, ST-CoSense unifies sampling and imputation with end-to-end gradient flow, while addressing the unique challenges of urban spatio-temporal sensing: graph-structured spatial dependencies, evolving temporal dynamics, and sequential budget-constrained sampling decisions.

B. Data Imputation

Research on data imputation spans a spectrum of methodologies that can be broadly grouped into three paradigms: (i) classical statistical methods, (ii) machine learning methods, and (iii) deep learning methods.

Classical statistical imputation methods use simple distributional properties of the data. A common example is mean imputation, which replaces missing entries with the arithmetic mean of the observed values. Another widely used technique is K-Nearest-Neighbor (KNN) imputation, which estimates a missing value from its most similar instances. Although KNN better preserves local correlations, its pairwise-distance calculations become computationally burdensome on large datasets.

Machine learning methods reconstruct missing entries by learning the latent structure and dependencies of the data. Matrix factorization approaches, such as Temporal Regularized Matrix Factorization (TRMF) [23], decompose data into low-rank representations to capture dominant patterns and fill missing values. Regression-based methods infer missing values by modeling relationships between observed and unobserved variables through techniques such as support vector regression and linear regression [24]. Ensemble methods, including random forests and gradient boosting machines, improve imputation accuracy by aggregating predictions from multiple weak learners [25], [26]. While these approaches demonstrate effectiveness on moderately-sized datasets, they are often limited by assumptions such as low-rank structure or linear relationships, constraining their ability to capture complex spatio-temporal dependencies in large-scale urban data.

To overcome these limitations, deep learning imputation methods have made significant progress in recent years by learning complex patterns through neural networks. These methods can be categorized by their primary modeling mechanisms. Recurrent methods [27] capture temporal dependencies through sequential processing with recurrent architectures that maintain hidden states throughout time steps. Convolutional methods [28], [29] extract temporal patterns efficiently using convolutions with local receptive fields. Attention-based

methods [30]–[34] leverage attention mechanisms to model long-range temporal dependencies by computing relationships across all time steps simultaneously. Generative methods [35]–[38] take a probabilistic approach, learning data distributions through diffusion models or GANs to generate plausible imputations. Graph-based methods [39]–[42] have gained prominence by explicitly modeling spatial relationships through graph structures, using graph convolution to propagate information between neighboring locations. As surveyed by Jin et al. [43], graph-based approaches are particularly effective for urban sensing scenarios where spatial topology guides the imputation at unobserved nodes.

For deep learning imputation methods, a common premise is that the primary learning signal comes from the spatio-temporal dependencies among observed locations, and that better reconstruction follows from modeling these dependencies more accurately. What remains largely unexplored, however, is the observation pattern itself. The observed locations are typically treated as an exogenous condition rather than a variable that can be optimized. As a result, reconstruction quality depends heavily on whether the selected observations are sufficiently informative, yet the model has no mechanism to assess their value or indicate where additional observations would be most beneficial. ST-CoSense addresses this by making the observation pattern itself a learnable component, jointly optimized with the imputation module so that sensing resources are actively directed toward the regions most critical for accurate recovery.

III. PRELIMINARY

In this section, we define some commonly used terms in ST-CoSense and explain our problem.

A. Definitions

Definition 1. Urban Graph. The urban target area is modeled as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ denotes the set of $|\mathcal{V}|$ monitoring subareas, each within the coverage range of a single sensor, and $\mathcal{E}$ represents the connections between these subareas. The associated adjacency matrix $\mathcal{A} \in \mathbb{R}^{|\mathcal{V}| \times |\mathcal{V}|}$ specifies pairwise connections according to the underlying spatial partitioning scheme.

Definition 2. Sampling Mask. Let $\vec{m}_t = \{m_t^{(1)}, m_t^{(2)}, \dots, m_t^{(|\mathcal{V}|)}\}$ be a binary vector where $m_t^{(i)} \in \{0, 1\}$ indicates whether the i -th subarea is sampled ($m_t^{(i)} = 1$) or not ($m_t^{(i)} = 0$) at time t . We denote the sequence of sampling patterns from time $t+1$ to $t+L$ as $\mathbf{M}_{t:t+L}$, which concatenates these binary vectors across the time window.

Definition 3. Urban Data. Let $\mathbf{X}_{t:t+L} = \{\vec{x}_{t+1}, \dots, \vec{x}_{t+L}\} \in \mathbb{R}^{|\mathcal{V}| \times L}$ denote the urban data matrix over a time window L starting from time $t+1$, where each $\vec{x}_t \in \mathbb{R}^{|\mathcal{V}|}$ contains data across $|\mathcal{V}|$ subareas. Only subareas marked as sampled in $\mathbf{M}_{t:t+L}$ contain valid observations, and others represent missing data.

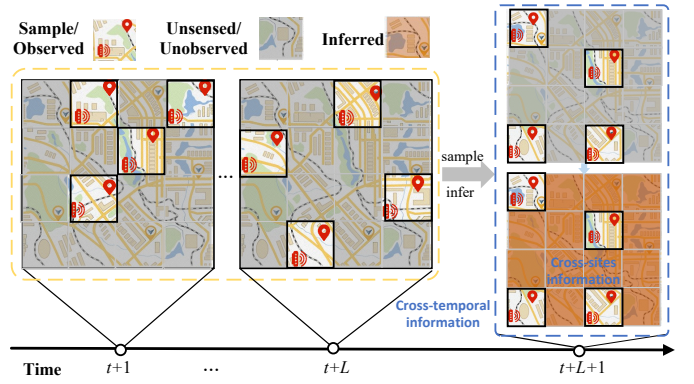


Fig. 2: Illustration of the urban sparse sensing problem. At each time slot, only a subset of subareas are observed under a budget constraint, while the remaining subareas are inferred via spatio-temporal correlations.

B. Scope and Premises

Before formally stating the optimization problem, we make explicit several premises about the sensing infrastructure and the monitored scenario, which clarify the applicable scope of ST-CoSense.

Dynamic Sensing. ST-CoSense selects a potentially different subset of subareas at each time step, presupposing that sensing locations can be reassigned across rounds. The time scale for each round typically ranges from minutes to hours, depending on the sensing platform and the monitored scenario. This aligns with the way many urban sensing systems operate in practice. In sparse mobile crowdsensing [10], [44], tasks are dispatched to different participants each round, so the set of covered locations varies naturally over time. Similar flexibility arises in UAV or vehicle-based monitoring [45], [46], where agents reposition between rounds under physical constraints. Our formulation abstracts away the specific reassignment mechanism, focusing on the location-selection decision itself rather than platform-level scheduling.

Uniform Sensing Cost. Following [7], [10], each subarea is treated as equally costly to sense, which simplifies the budget constraint to a cap on the number of sampled locations per step. This simplification isolates the where-to-sample problem from platform-specific cost considerations. In settings with heterogeneous sensing costs, the cardinality constraint can be replaced by a knapsack-style budget without architectural modifications.

Spatio-Temporal Correlation. The imputation module infers missing values from spatial and temporal dependencies, which requires such structure to be present in the data. In practice, urban sensing targets generally satisfy this condition. For example, traffic flows exhibit strong spatial autocorrelation along road networks and clear temporal periodicity tied to commuting patterns [47].

Under these premises, we now formally define the urban sparse sensing problem.

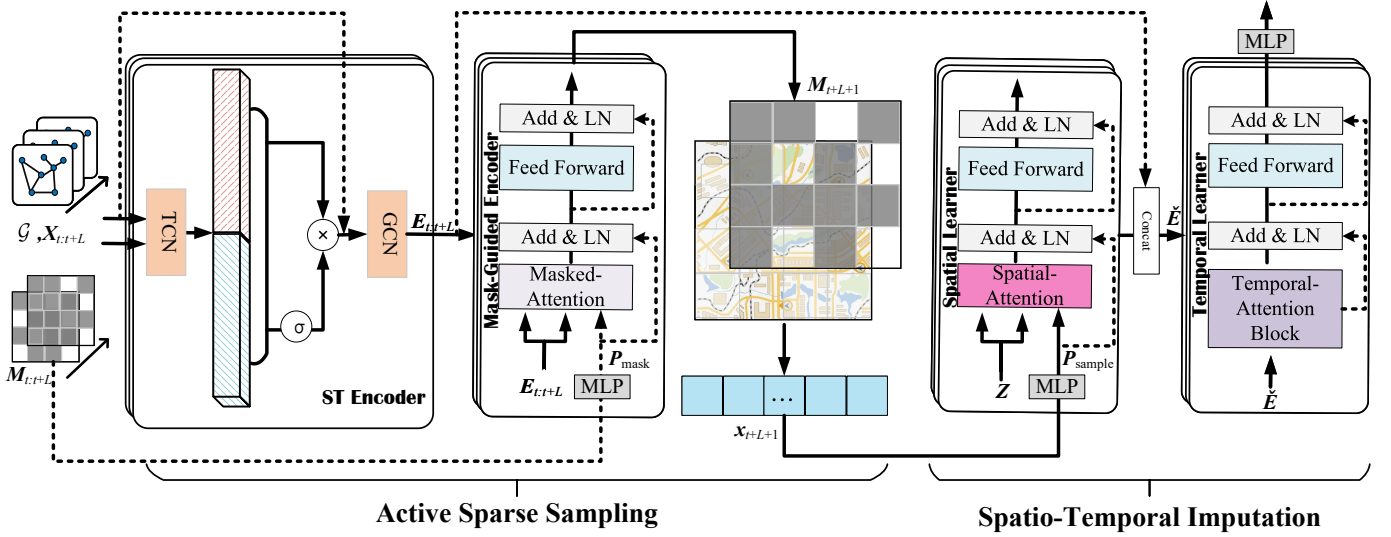


Fig. 3: Overall framework of ST-CoSense, comprising active sparse sampling and spatio-temporal imputation modules.

C. Problem Statement

The urban sparse sensing problem can be stated as follows. Given a limited sampling budget, select which subareas to observe at each time step and reconstruct the complete data from these sparse observations. Fig. 2 illustrates this setup with a concrete example. The target urban area is partitioned into a 4×4 grid of subareas. At each time slot, budget constraints limit sampling to a small subset. For example, only four out of 16 subareas are sampled from time slots $t+1$ to $t+L+1$, with the exact number determined by the available sensing budget.

Our framework operates as follows: observations collected over L historical time slots inform the selection of subareas to monitor at time slot $t+L+1$. The spatio-temporal imputation module then exploits correlations across the $(L+1)$ -slot window to reconstruct values for unsensed subareas, enabling accurate fine-grained urban sensing despite limited sampling resources.

Formally, given historical observed spatio-temporal data $\mathbf{X}_{t:t+L} \odot \mathbf{M}_{t:t+L}$ (the symbol $\odot$ denotes element-wise multiplication) and urban graph $\mathcal{G}$, we aim to determine the optimal subset of subareas for the next time slot that minimize imputation error while satisfying the budget constraint. This optimization can be formulated as:

$$\mathcal{F}(\mathbf{X}_{t:t+L}, \mathbf{M}_{t:t+L}, \mathcal{G}, \alpha) \rightarrow [\mathbf{M}_{t+L+1}, \hat{\mathbf{X}}_{t+L+1}], \quad (1)$$

where $\mathcal{F}$ is the mapping function (ST-CoSense), α denotes the budget constraint on sensor deployment, and $\hat{\mathbf{X}}_{t+L+1}$ represents the imputed result.

IV. METHODOLOGY

This section details the **ST-CoSense** framework. As illustrated in Fig. 3, the framework comprises two tightly coupled modules: (i) the *Active Sparse Sampling Module*, which determines optimal sampling locations under budget constraints, and (ii) the *Spatio-Temporal Imputation Module*, which reconstructs complete urban data from sparse observations.

A. Active Sparse Sampling Module

The active sparse sampling module consists of two components: a gated Spatio-Temporal (ST) encoder that extracts latent representations from historical observations, and a mask-guided encoder that fuses these representations with historical sampling patterns to produce an informativeness score for each subarea. The top-scoring subareas are selected for sensing at the next time step.

1) *Spatio-Temporal Encoder*: Inspired by the gating mechanism in [48], we design a gated ST encoder comprising multiple stacked blocks. Each block processes temporal and spatial information through two sequential stages, allowing temporal and spatial dependencies to be modeled independently.

In the first step, observed data $\mathbf{X}_{t:t+L} \odot \mathbf{M}_{t:t+L}$ is projected into a hidden state, denoted $\mathbf{X} \odot \mathbf{M}$ for brevity. The Temporal Convolutional Network (TCN) [49] is employed to model long-range temporal dependencies from $\mathbf{X} \odot \mathbf{M}$. The TCN output is split into two branches along the feature dimension:

$$\text{TCN}(\mathbf{X} \odot \mathbf{M}) = [\text{TCN}(\mathbf{X} \odot \mathbf{M})_{\text{up}}, \text{TCN}(\mathbf{X} \odot \mathbf{M})_{\text{down}}], \quad (2)$$

where $\text{TCN}(\mathbf{X} \odot \mathbf{M})_{\text{up}}$ captures salient temporal features and $\text{TCN}(\mathbf{X} \odot \mathbf{M})_{\text{down}}$ serves as a gating term controlling the influence of the learned features. These embeddings are integrated via sigmoid gating $\sigma(\cdot)$ with residual connections:

$$\mathbf{E}_{t:t+L} = \text{TCN}(\mathbf{X} \odot \mathbf{M})_{\text{up}} \odot \sigma(\text{TCN}(\mathbf{X} \odot \mathbf{M})_{\text{down}}) + \mathbf{X} \odot \mathbf{M}. \quad (3)$$

The sigmoid function enables element-wise control over how much temporal information is retained. The gating mechanism selectively retains salient temporal features, while the residual connection preserves the original input signal to ensure stable gradient flow.

In the second step, spatial dependencies across subareas are modeled by applying Graph Convolutional Networks (GCN) [50] to the temporal embeddings obtained in the first step.

Given the urban graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, GCN aggregates information from neighboring subareas:

$$\mathbf{E}_{t:t+L} = \text{GCN}(\mathbf{E}_{t:t+L}, \mathcal{G}). \quad (4)$$

By stacking multiple ST blocks, the encoder progressively aggregates temporal and spatial information. Hierarchical temporal aggregation then compresses the temporal dimension from L steps to a single vector per subarea:

$$\mathbf{E}_{t:t+L} \in \mathbb{R}^{|\mathcal{V}| \times 1 \times D}, \quad (5)$$

where D denotes the embedding dimensionality. This compression acts as an information bottleneck, forcing the encoder to distill the most discriminative temporal characteristics for each subarea rather than retaining all sequential detail. In practice, adjacent time steps in urban sensing data tend to be highly correlated, so much of the raw temporal signal is redundant. Aggregating across the window retains the essential dynamics while discarding this redundancy, and simultaneously reduces the attention cost of downstream spatial operations from $\mathcal{O}((|\mathcal{V}|L)^2)$ to $\mathcal{O}(|\mathcal{V}|^2)$.

Remark 1. (Complexity Analysis of ST Encoder). Each block processes data through two sequential operations. The TCN processes each subarea's temporal sequence independently, requiring $\mathcal{O}(|\mathcal{V}|L)$ floating-point operations across all subareas. The K_g -order Chebyshev Graph Convolution aggregates spatial information at each time step, requiring $\mathcal{O}(K_g|\mathcal{E}|L) \approx \mathcal{O}(|\mathcal{E}|L)$ operations (since K_g is a small constant). The total per-block cost is therefore $\mathcal{O}((|\mathcal{V}| + |\mathcal{E}|)L)$, scaling linearly in both the number of subareas and the time window length.

2) *Mask-Guided Encoder:* While the ST encoder effectively integrates spatio-temporal features, relying solely on such features neglects the influence of historical sampling patterns. Historical sampling decisions inherently contain valuable information that can significantly influence future sampling strategies. Therefore, to optimally exploit historical sampling knowledge, a mask-guided encoder is designed to dynamically combine historical sampling masks with learned spatio-temporal representations, thus optimizing sampling decisions.

The historical sampling mask matrix $\mathbf{M}_{t:t+L} \in \mathbb{R}^{|\mathcal{V}| \times L}$ is transformed into a high-dimensional embedding space, matching the dimensionality of the spatio-temporal embeddings $\mathbf{E}_{t:t+L}$:

$$\mathbf{P}_{\text{mask}} = \text{MLP}(\mathbf{M}_{t:t+L}) \in \mathbb{R}^{|\mathcal{V}| \times D}, \quad (6)$$

where the Multi-Layer Perceptron (MLP) projects historical binary sampling masks into continuous semantic representations. The historical masks encode past sensing distributions, indicating which subareas have been frequently or rarely sampled, thereby complementing spatio-temporal embeddings to reduce redundant sampling and improve spatial coverage.

Although the masks themselves do not directly encode imputation errors, their influence on reconstruction quality is learned implicitly through end-to-end training. During optimization, gradients propagate through the mask-guided encoder, enabling the model to associate historical sampling patterns with downstream imputation performance.

A masked-attention mechanism then integrates mask embeddings with spatio-temporal features. Specifically, the masked-attention mechanism employs the historical mask embedding as the query vector, enabling it to dynamically weigh the importance of historical spatio-temporal patterns across subareas. The resulting embedding is further refined through residual connections, Layer Normalization (Norm), and a Feed-Forward Neural network (FFN), enhancing model robustness and representation quality:

$$\begin{aligned} \mathbf{E}_{\text{sample}}^{(0)} &= \text{Attn}(\mathbf{P}_{\text{mask}}, \mathbf{E}_{t:t+L}, \mathbf{E}_{t:t+L}), \\ \mathbf{E}_{\text{sample}}^{(1)} &= \text{Norm}(\mathbf{E}_{\text{sample}}^{(0)} + \mathbf{P}_{\text{mask}}), \\ \mathbf{E}_{\text{sample}}^{(2)} &= \text{Norm}(\mathbf{E}_{\text{sample}}^{(1)} + \text{FFN}(\mathbf{E}_{\text{sample}}^{(1)})). \end{aligned} \quad (7)$$

Leveraging the refined embedding $\mathbf{E}_{\text{sample}}^{(2)}$ and a predefined sampling budget constraint α , we select the top- α subareas for sampling at the subsequent time slot $t + L + 1$:

$$\begin{aligned} \mathbf{M}_{t+L+1} &= \text{topK}(s, \alpha), \\ s &= \text{MLP}(\mathbf{E}_{\text{sample}}^{(2)}) \in \mathbb{R}^{|\mathcal{V}|}. \end{aligned} \quad (8)$$

where $\text{topK}(s, \alpha) : \mathbb{R}^{|\mathcal{V}|} \rightarrow \{0, 1\}^{|\mathcal{V}|}$ selects the α subareas with highest scores and sets their corresponding mask values to 1.

By explicitly modeling historical sampling patterns through learnable embeddings and attention-based fusion, the mask-guided encoder reduces sampling redundancy and improves coverage efficiency. This enables prioritization of under-sampled yet informative locations, enhancing overall imputation quality through end-to-end optimization.

Remark 2. (Complexity Analysis of Mask-Guided Encoder). The mask-guided encoder contains several computational blocks: MLP, the attention mechanism, and FFN, but the dominant cost arises from the attention itself. Computing the query-key dot products and aggregating the values incurs $\mathcal{O}(|\mathcal{V}|^2)$ floating-point operations. Although this complexity is quadratic in $|\mathcal{V}|$, it completely eliminates the temporal factor L that inflates the cost of full spatio-temporal self-attention, reducing the complexity from $\mathcal{O}((|\mathcal{V}|L)^2)$ to quadratic in $|\mathcal{V}|$.

B. Spatio-Temporal Imputation

The imputation module comprises spatio-temporal learners integrated with attention mechanisms designed to precisely infer missing values for unsensed subareas. These learners systematically identify patterns in current observations and integrate them with historical insights to capture complex spatio-temporal dependencies, generating accurate imputations through iterative refinement.

1) *Spatial Learner:* The spatial learner models spatial relationships through an attention mechanism operating on two complementary embeddings: a learnable subarea embedding $\mathbf{Z} \in \mathbb{R}^{|\mathcal{V}| \times 1 \times D}$, which is progressively refined during training to encode persistent spatial correlations (e.g., which locations typically influence each other), and a sampling embedding $\mathbf{P}_{\text{sample}} \in \mathbb{R}^{|\mathcal{V}| \times 1 \times D}$ derived from the current observations $\mathbf{X}_{t+L+1} \odot \mathbf{M}_{t+L+1}$, which captures the real-time observational

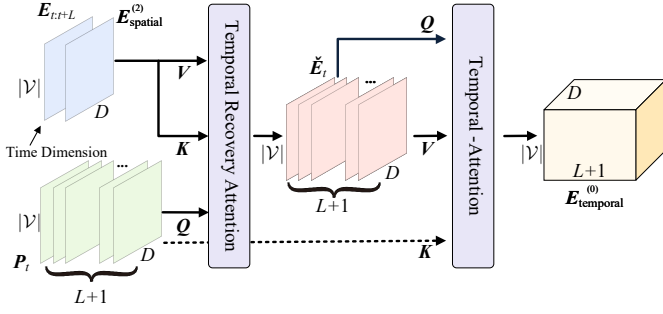


Fig. 4: The Temporal-Attention Block. It primarily consists of two mechanisms: the Temporal Recovery Attention and the Temporal-Attention.

state. The complete spatial learner process can be expressed as:

$$\begin{aligned} \mathbf{E}_{\text{spatial}}^{(0)} &= \text{Attn}(\mathbf{Z}, \mathbf{Z}, \mathbf{P}_{\text{sample}}), \\ \mathbf{E}_{\text{spatial}}^{(1)} &= \text{Norm}(\mathbf{E}_{\text{spatial}}^{(0)} + \mathbf{P}_{\text{sample}}), \\ \mathbf{E}_{\text{spatial}}^{(2)} &= \text{Norm}(\mathbf{E}_{\text{spatial}}^{(1)} + \text{FFN}(\mathbf{E}_{\text{spatial}}^{(1)})). \end{aligned} \quad (9)$$

This design adopts the learned-query structural aggregation paradigm [51]. The learnable embeddings $\mathbf{Z}$ serve as Q and K to define a structural relationship matrix, while the actual observational data $\mathbf{P}_{\text{sample}}$ in V provides the content to be aggregated. The attention weights thus reflect the spatial structure learned from training, while the aggregated output adapts to transient observational conditions at each time step. This separation lets the model exploit learned spatial structure while remaining sensitive to the specific observations available at each time step.

Remark 3. (Complexity Analysis of Spatial Learner). Similar to the mask-guided encoder, the main computational burden of the spatial learner stems from the attention module. With $|\mathcal{V}|$ subarea tokens, the query-key dot products and value aggregation require $\mathcal{O}(|\mathcal{V}|^2)$ floating-point operations.

2) *Temporal Learner:* While the spatial learner captures inter-location dependencies, it cannot model sequential dynamics along the historical sequence. To address this, the temporal-attention block is designed to model complex temporal dependencies (as shown in Fig. 4).

The spatial embedding $\mathbf{E}_{\text{spatial}}^{(2)}$ and the historical data representation $\mathbf{E}_{t:t+L}$ are concatenated to form a representation $\tilde{\mathbf{E}} \in \mathbb{R}^{|\mathcal{V}| \times 2 \times D}$:

$$\tilde{\mathbf{E}} = \text{Concat}(\mathbf{E}_{t:t+L}, \mathbf{E}_{\text{spatial}}^{(2)}, \dim = 1). \quad (10)$$

Considering that the temporal dimension of $\mathbf{E}_{t:t+L}$ has been compressed in previous stages, inspired by [52], [53], we employ a learnable temporal projection vector $\mathbf{P}_t \in \mathbb{R}^{|\mathcal{V}| \times (L+1) \times D}$ to restore the original temporal granularity. Specifically, we apply a temporal recovery attention mechanism:

$$\tilde{\mathbf{E}}_t = \text{Attn}(\mathbf{P}_t, \tilde{\mathbf{E}}, \tilde{\mathbf{E}}), \quad (11)$$

where $\tilde{\mathbf{E}}_t \in \mathbb{R}^{|\mathcal{V}| \times (L+1) \times D}$ is the reconstructed representation. The learnable temporal projection vector $\mathbf{P}_t$ acts as probes,

each learning to extract the information corresponding to a specific temporal position from the compressed representation.

To further refine temporal patterns and effectively weight temporal contexts, we implement another attention mechanism:

$$\mathbf{E}_{\text{temporal}}^{(0)} = \text{Attn}(\tilde{\mathbf{E}}_t, \mathbf{P}_t, \tilde{\mathbf{E}}_t). \quad (12)$$

By placing $\tilde{\mathbf{E}}_t$ as Q and $\mathbf{P}_t$ as K, each recovered time step dynamically determines its temporal attention pattern based on its own data content, oriented against the positional coordinate system provided by $\mathbf{P}_t$, while $V = \tilde{\mathbf{E}}_t$ ensures the output carries recovered features. Together with Eq. (11), the two mechanisms form a complementary pair, ensuring bidirectional interaction between learned temporal structure and actual data content [53], [54].

The resulting embedding is then refined through a residual FFN block with Norm to improve its representational capacity:

$$\begin{aligned} \mathbf{E}_{\text{temporal}}^{(1)} &= \text{Norm}(\mathbf{E}_{\text{temporal}}^{(0)} + \tilde{\mathbf{E}}_t), \\ \mathbf{E}_{\text{temporal}}^{(2)} &= \text{Norm}(\mathbf{E}_{\text{temporal}}^{(1)} + \text{FFN}(\mathbf{E}_{\text{temporal}}^{(1)})), \end{aligned} \quad (13)$$

where $\mathbf{E}_{\text{temporal}}^{(2)}$ captures temporal patterns. The learnable vector $\mathbf{P}_t$ plays a dual role in this process. In Eq. (11), it serves as temporal probes to restore the compressed temporal dimensionality. In Eq. (12), it serves as a temporal coordinate system providing positional anchors for content-adaptive temporal refinement.

Our temporal learner addresses the critical issue of temporal compression by integrating a learnable temporal projection and attention-based temporal reconstruction. This design choice enables historical temporal information to be adaptively leveraged, improving the model's capacity to handle dynamic temporal variations.

Remark 4. (Complexity Analysis of Temporal Learner). The temporal learner executes two cross-attention mechanisms in sequence. (i) Temporal Recovery attention: For each subarea, the $(L+1)$ query tokens attend to the two key-value tokens, requiring $(L+1) \times 2$ dot products per subarea. (ii) Temporal attention: $(L+1)$ query tokens attend to $(L+1)$ key tokens, requiring $(L+1)^2$ dot products per subarea. Summing the two blocks yields $\mathcal{O}(|\mathcal{V}|L^2)$ operations. This cost is substantially lower than the $\mathcal{O}(|\mathcal{V}|L^2)$ complexity of full spatio-temporal self-attention, as the temporal learner processes each subarea's temporal sequence independently rather than jointly attending over all $|\mathcal{V}| \times L$ tokens.

The spatio-temporal learning process decomposes the underlying data structure into spatial and temporal components, which are modeled by dedicated learners. The spatial learner captures spatial correlations by modeling inter-subarea dependencies. Meanwhile, the temporal learner extracts sequential dependencies by identifying patterns across time slots. Finally, MLP is used to output the imputed results.

C. Sampling and Imputation Process

Training. ST-CoSense is trained end-to-end in a closed loop in which the imputed signal itself acts as supervision for the sampler. At each update step, the gated ST encoder

and the mask-guided encoder propose an informativeness score for each subarea. A differentiable relaxation converts these scores into a budget-constrained mask. The spatio-temporal learner then reconstructs the missing values $\hat{\mathbf{X}}_{t+L+1}$ from the observations selected by this mask. The loss function is the reconstruction error:

$$\mathcal{L}_{\text{rec}} = \frac{1}{|\mathcal{V}|} \|\hat{\mathbf{X}}_{t+L+1} - \mathbf{X}_{t+L+1}\|_1. \quad (14)$$

$\mathcal{L}_{\text{rec}}$ is back-propagated through both the imputer and the sampler. The gradient $\partial \mathcal{L}_{\text{rec}} / \partial \mathbf{E}_{\text{sample}}^{(2)}$ indicates the sensitivity of the reconstruction error to each subarea's embedding, guiding the sampling policy toward locations that most reduce imputation error. Subareas whose inclusion significantly reduces imputation error receive large gradients, driving parameter updates that increase the likelihood of selecting those locations in future steps. Conversely, redundant locations that contribute minimally to error reduction receive small gradients and are gradually deprioritized. This gradient-based feedback enables the sampling policy to learn which locations are most informative to minimize imputation error.

Differentiable relaxation (e.g., Gumbel-Softmax or Straight-Through Estimator) serves only to provide a gradient pathway through the discrete top-K selection in Eq. (8). Since the top-K decision depends only on the score ranking rather than absolute magnitudes, training is insensitive to the specific relaxation scheme, and reasonable relaxation does not alter the result of selection.

Inference. During deployment, the model operates in a rolling-window fashion. For each new window, it generates a mask for the next step using the learned policy, acquires the corresponding measurements, and merges them with historical context to produce the city-wide estimate $\hat{\mathbf{X}}_{t+L+1}$. As the window advances, new observations replace older ones, and the sampling policy adapts to the evolving spatio-temporal context.

V. EXPERIMENTS

In this section, we evaluate ST-CoSense on multiple real-world datasets to answer the following questions:

- **RQ1:** Does ST-CoSense achieve state-of-the-art sensing performance compared to existing baselines?
- **RQ2:** How does each sub-module contribute, and do the performance gains stem from sampling-imputation coupling?
- **RQ3:** How robust is ST-CoSense in different settings?
- **RQ4:** How does ST-CoSense perform in terms of temporal and spatial dimensions?
- **RQ5:** What is the training efficiency of ST-CoSense?

A. Experimental Settings

1) *Datasets:* In the experiments, we mainly use four real-world datasets: NYCBike [55], AQI [56], METR-LA [11], and a large-scale urban dataset GLA [57]. The descriptions related to the datasets are shown in Tab. I.

Graph construction follows directly from the spatial meta-information provided by each data source. For GLA, we

TABLE I: Description and Statistics of Datasets

Dataset	NYCBike	AQI	METR-LA	GLA
Data Type	Bike Rent	PM2.5	Traffic Speed	Traffic Flow
Location	NY, US	BJ,etc, CHN	LA, US	Greater LA, US
Subareas	128	437	207	3834
Steps	4392	8760	34272	105120
Time Interval	60min	60 min	5 min	5 min
Time Span	04/01/2014-09/30/2014	05/01/2014-04/30/2015	03/01/2012-06/27/2012	01/01/2019-12/31/2019

use the pre-constructed adjacency matrix released as part of the LargeST benchmark. For NYCBike, connectivity is defined by 8-neighborhood adjacency on the 16x8 spatial grid, consistent with prior works [58], [59]. For AQI and METR-LA, both datasets provide sensor coordinates, and adjacency relationships are computed via a thresholded Gaussian kernel.

2) *Baselines:* To evaluate the performance of ST-CoSense, we compare it with classic models and state-of-the-art methods for spatio-temporal sensing. The baselines include: (i) **Traditional methods:** MEAN and KNN, which rely on simple statistical aggregation, and TRMF [23], a matrix factorization approach; (ii) **Sampling-based methods:** AS-MCS [7] and OS-MCS [10], which optimize sampling strategies; and (iii) **Deep learning methods:** including recurrent methods (BRITS [27]), convolutional methods (TimesNet [28], ModernTCN [29]), attention-based methods (Nonstationary Transformer [30], Crossformer [31], iTransformer [32], Imputeformer [33], SAITS [34]), generative models (US-GAN [36], CSDI [35]), and graph-based methods (GRIN [39], StemGNN [40], GSLI [42]).

3) *Parameter Settings:* All datasets are partitioned chronologically at a 7:1:2 ratio (train/validation/test), with strict temporal boundaries ensuring that no future information leaks into training. Since the test period is entirely unseen during training and may exhibit different temporal patterns, this setup effectively evaluates the model's ability to generalize under temporal distribution shift. Training batches are randomly shuffled within the training set to improve optimization stability.

The time window size is set to 24, with the convolution kernel size and graph neighborhood order both set to 3, and the embedding dimension set to 64. The model is trained using the ADAM optimizer for 200 epochs, with a learning rate of 0.001 and a batch size of 32. The MLP adopts a 2-layer fully-connected architecture with a hidden dimension of 64 and ReLU activation. The primary modeling capacity of the framework is concentrated in the attention mechanisms, and the MLP intentionally serves as a lightweight projection module with its hidden dimension matching the embedding dimension across the framework.

B. Overall Performance (RQ1)

We evaluate ST-CoSense at a 50% sampling rate, where half of the subareas are observed at each time step. This setting balances sensing cost reduction with sufficient information for

TABLE II: Comparison of various models on NYCBike, AQI, and METR-LA datasets under 50% sampling rate scenario.

Model	NYCBike		AQI		METR-LA		GLA	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
MEAN	7.46±0.00	15.08±0.00	34.12±0.00	52.38±0.00	7.55±0.00	11.93±0.00	144.85±0.00	166.69±0.00
KNN	7.30±0.00	13.37±0.00	24.62±0.00	41.35±0.00	7.98±0.00	12.02±0.00	-	-
TRMF	4.66±0.02	8.24±0.09	15.50±0.11	26.37±0.36	4.90±0.02	7.41±0.05	-	-
AS-MCS	3.69±0.01	5.54±0.05	15.14±0.01	26.83±0.04	3.76±0.03	4.84±0.06	-	-
OS-MCS	3.31±0.01	5.15±0.03	14.83±0.01	23.20±0.02	3.42±0.09	4.82±0.17	-	-
Brits	3.94±0.02	6.19±0.08	16.60±0.16	26.64±2.53	2.57±0.00	4.49±0.02	32.95±1.20	52.06±2.13
TimesNet	2.96±0.01	6.15±0.02	13.33±0.03	24.76±0.12	3.07±0.01	6.20±0.06	N.A.	N.A.
ModernTCN	2.37±0.06	4.73±0.11	11.99±0.01	22.75±0.09	2.95±0.02	5.91±0.04	N.A.	N.A.
Nonstationary Transformer	2.99±0.04	6.21±0.22	12.89±0.01	23.99±0.06	2.94±0.09	5.92±0.18	N.A.	N.A.
Crossformer	3.17±0.03	6.85±0.13	15.24±0.01	26.68±0.06	3.92±0.01	6.43±0.03	22.90±0.29	36.70±0.45
iTransformer	3.34±0.04	7.20±0.13	13.10±0.13	24.28±0.20	2.99±0.03	5.55±0.11	OOM	OOM
Imputeformer	2.08±0.01	4.37±0.03	11.70±0.02	23.39±0.08	2.42±0.05	5.27±0.07	19.29±0.22	32.75±0.59
SAITS	2.25±0.13	4.74±0.31	15.79±0.42	27.14±0.30	3.76±0.46	8.20±1.19	25.94±0.21	47.05±0.58
US-GAN	3.44±0.01	5.47±0.06	16.33±0.04	27.03±0.07	4.72±0.08	7.95±0.17	29.44±2.04	48.12±2.25
CSDI	2.46±0.02	5.15±0.06	11.14±0.05	22.23±0.31	2.71±0.03	5.13±0.08	23.36±0.64	37.79±0.15
StemGNN	2.28±0.04	4.75±0.12	16.45±0.09	27.45±0.46	4.64±0.24	9.38±0.55	26.91±0.38	47.80±0.39
GSLI	2.08±0.01	4.28±0.06	15.07±0.31	25.94±0.54	2.44±0.09	4.67±0.12	N.A.	N.A.
GRIN	2.06±0.02	4.23±0.07	11.89±0.33	23.40±0.57	2.03±0.04	3.47±0.78	17.70±0.08	30.53±0.76
ST-CoSense	1.94±0.02	3.90±0.08	10.13±0.13	19.89±0.32	1.98±0.03	3.15±0.12	17.74±0.06	29.02±0.20

imputation and aligns with configurations commonly used in prior work, ensuring fair comparison. We additionally evaluate on the large-scale GLA dataset to test scalability, where only competitive baselines are included.

The experimental results are summarized in Tab. II. Overall, ST-CoSense shows strong performance across all four datasets. On NYCBike, it achieves MAE of 1.94 and RMSE of 3.90, representing 5.82% and 7.80% improvements respectively, compared with the best-performing baseline. Similar gains are observed on AQI, where ST-CoSense achieves MAE of 10.13 and RMSE of 19.89, 9.07% and 10.53% better than the best-performing baseline. On METR-LA, ST-CoSense outperforms the best-performing baseline by 2.46% and 9.22% respectively. These consistent improvements across datasets with different characteristics demonstrate that the joint optimization of sampling and imputation generalizes across sensing modalities and temporal resolutions. On the large-scale GLA dataset with 3,834 nodes, ST-CoSense achieves MAE of 17.74, nearly matching GRIN's 17.70 (a difference of 0.04), while achieving a 4.95% RMSE reduction relative to it. This indicates greater robustness to large errors in city-scale deployments.

Traditional methods such as MEAN and KNN exhibit consistently poor performance across all datasets, primarily due to their simplistic imputation strategies that neglect spatio-temporal characteristics. On datasets like AQI, which exhibits high spatio-temporal dynamics, these methods fail to capture intricate relationships, resulting in poor imputation performance. Matrix factorization-based methods, such as TRMF, partially address temporal dependencies but are constrained by low-rank assumptions, limiting their ability to capture high-order correlations in real-world spatio-temporal data. Similarly, methods such as AS-MCS and OS-MCS, despite optimizing the sampling process, still rely on matrix factorization for imputation. While these methods show improvement over traditional approaches, their performance remains suboptimal due to the inherent limitations of matrix factorization. Deep learning approaches, such as CSDI, GSLI, Imputeformer, and GRIN, generally outperform traditional and matrix factorization methods across all datasets. These

models leverage advanced architectures, such as multi-layer neural networks and attention mechanisms, to capture complex spatio-temporal dependencies. However, under low sampling rates, the scarcity of effective data poses significant challenges to the models' generalization and robustness.

ST-CoSense overcomes these limitations through joint optimization of sampling and imputation, achieving consistent improvements across datasets with diverse characteristics, sensing modalities, and temporal resolutions.

C. Ablation Study (RQ2)

To further analyze the contribution of each sub-module in ST-CoSense, we conduct an ablation study by comparing it with its variants, as detailed in Tab. III. The variants include:

- *w/o spatial learner*: remove the spatial dependencies learning in Eq.(9).
- *w/o temporal learner*: remove the temporal dependencies learning component in Eq.(10) - Eq.(13).
- *w/o mask-guided encoder*: remove the mask-guided encoder in Eq.(6)-Eq.(8). The sampling strategy degrades to uniform random selection. The ST encoder and imputation module remain unchanged, isolating the contribution of mask-guided informed sampling.
- *w/o Z*: replace **Z** with a fixed spatial position embedding in Eq.(9).

TABLE III: Ablation studies on ST-CoSense.

Variation	NYCBike		AQI		METR-LA	
	MAE	RMSE	MAE	RMSE	MAE	RMSE
ST-CoSense	1.94	3.90	10.13	19.89	1.98	3.15
w/o mask-guided encoder	2.29	4.52	12.54	23.21	2.65	4.13
w/o spatial learner	2.23	4.44	10.61	20.50	2.75	4.55
w/o temporal learner	2.57	5.13	12.07	22.41	2.66	4.13
w/o Z	2.12	4.18	10.34	20.21	2.42	4.05
Imputeformer+Sampling	2.03	4.26	11.42	22.95	2.36	5.16
GRIN+Sampling	2.01	4.12	11.61	22.98	2.00	3.38
StemGNN+Sampling	2.21	4.60	16.12	27.06	4.52	9.17
GSLI+Sampling	2.04	4.13	14.87	25.22	2.36	4.51

The experimental results in Tab. III demonstrate the necessity of each module. Removing the mask-guided encoder

and replacing it with random sampling increases MAE by 18.0% on NYCBike, 23.8% on AQI, and 33.8% on METR-LA, confirming the critical role of active sampling in directing sensing resources toward informative locations. Notably, even without the sampling module, ST-CoSense still outperforms most baselines (e.g., on METR-LA, MAE of 2.65 vs. ModernTCN’s 2.95), demonstrating that the imputation module alone captures complex dependencies effectively. Within the imputation module, removing the spatial learner increases MAE from 1.94 to 2.23 on NYCBike (14.9% degradation), while removing the temporal learner increases MAE to 2.57 (32.5% degradation), indicating that both sub-modules are critical for accurate imputation.

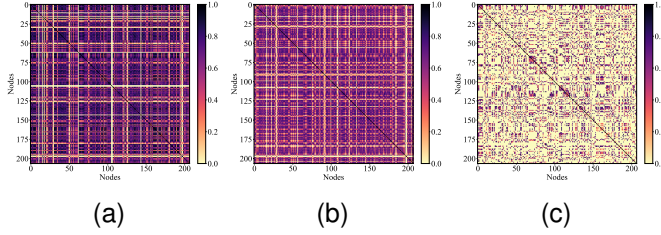


Fig. 5: Similarity matrices on the METR-LA. (a) Data-driven Similarity; (b) Subarea Embedding Similarity; (c) Graph-based Similarity.

Having confirmed the necessity of each module, we further examine whether the specific design choices within the spatio-temporal learners are well-founded. Fig. 5 visualizes three pairwise node similarity matrices on METR-LA. The subarea embedding similarity in (b) recovers the broad structure seen in (a), confirming alignment with the underlying data patterns. It also assigns meaningful similarity to node pairs that are distant in the road network yet share recurrent traffic dynamics, a relational pattern the fixed graph in (c) cannot represent. This gap between (b) and (c) demonstrates the effectiveness of subarea embedding $\mathbf{Z}$. Quantitatively, replacing the learnable $\mathbf{Z}$ with a fixed spatial position embedding (w/o $\mathbf{Z}$ in Tab. III) increases MAE by 9.3% on NYCBike and 22.2% on METR-LA, confirming that the learned relational structure contributes substantially beyond fixed positional information. We also verify the Q/K/V allocation strategy: changing $\text{Attn}(\mathbf{Z}, \mathbf{Z}, \mathbf{P}_{\text{sample}})$ to $\text{Attn}(\mathbf{Z}, \mathbf{P}_{\text{sample}}, \mathbf{P}_{\text{sample}})$ in the Spatial Learner raises MAE from 1.94 to 2.14 on NYCBike and from 10.13 to 10.59 on AQI, while changing $\text{Attn}(\tilde{\mathbf{E}}_t, \mathbf{P}_t, \tilde{\mathbf{E}}_t)$ to $\text{Attn}(\mathbf{P}_t, \mathbf{P}_t, \tilde{\mathbf{E}}_t)$ in the Temporal Learner raises MAE to 2.09 and 10.41 respectively. These results validate that the allocations follow the design principles discussed in Section IV.

The ablation above confirms the necessity of each module. A natural follow-up question is whether the performance advantage arises from the co-designed architecture or merely from additional computational capacity introduced by the sampling module. Therefore, we plug the Active Sparse Sampling module into different types of baselines. As shown in Tab. III, all plug-in variants improve over their respective baselines, yet consistently fall short of ST-CoSense. We attribute this to the tight architectural co-design in ST-CoSense: the sampling and imputation modules share a unified feature space and

are optimized toward a consistent objective, whereas plug-in combinations introduce representational mismatch between independently designed components.

D. Robustness Analysis (RQ3)

We assess the robustness of ST-CoSense by comparing it with four high-performing baselines (CSDI, GRIN, GSLI, Imputeformer) and one variant (namely w/o mask-guided encoder, and for simplicity, w/o MGE is used) under varying sampling rates.

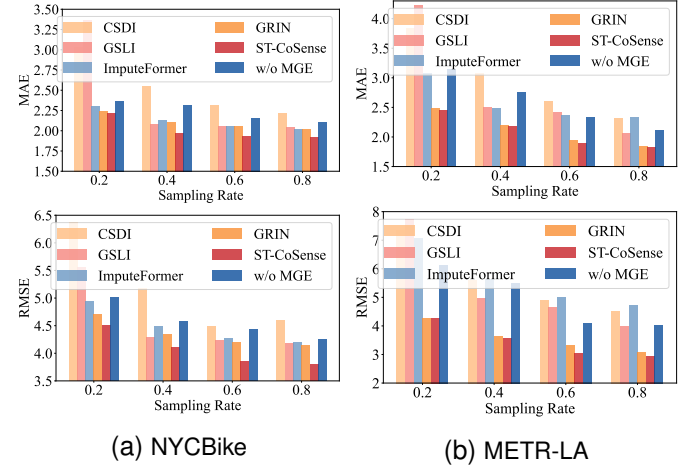


Fig. 6: The performance under different sampling rates.

As shown in Fig. 6, ST-CoSense consistently outperforms all baselines across the entire range. At low sampling rates (e.g., 20%), where limited observations make imputation particularly challenging, ST-CoSense maintains a clear advantage, demonstrating the effectiveness of the active sampling module in selecting informative locations under tight budget constraints. As the sampling rate increases, performance gaps among methods narrow, but ST-CoSense retains the lead. Among the baselines, GRIN and Imputeformer exhibit relatively strong performance owing to their fine-grained spatio-temporal dependency modeling, while GSLI shows competitive results at higher sampling rates but degrades more noticeably under sparse conditions.

A key observation is that ST-CoSense achieves competitive performance even at low sampling rates, a capability largely enabled by the active sparse sampling module. By ensuring coverage of critical data regions, the active sparse sampling mechanism allows the model to infer the missing value with high accuracy even under severe data scarcity. The effectiveness of the active sparse sampling module is further supported by the singular value distribution analysis presented in Fig. 7. Here, r is the number of singular values, σ_i denotes the i -th singular value, and $g(r)$ represents the proportion of the top r singular values relative to the total, as defined by:

$$g(r) = \frac{\sum_{i=1}^r \sigma_i^2}{\sum_{i=1}^n \sigma_i^2}. \quad (15)$$

A higher concentration of variance in the top singular values indicates that the selected observations span a more informative subspace, confirming that the active sampling module

prioritizes locations that collectively capture the principal structure of the urban field.

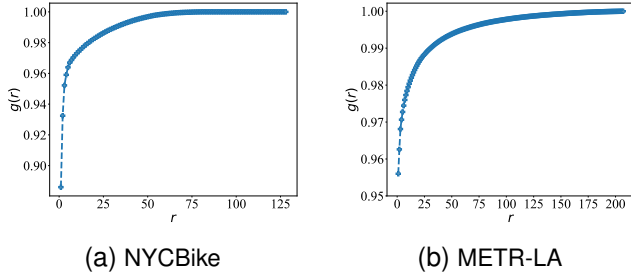


Fig. 7: The proportion of the total variance occupied by the top r singular values.

Beyond robustness to missing patterns, we further investigate the cross-rate generalization capability of ST-CoSense. This is of practical importance, as the available sensing budget may vary across deployment scenarios. To this end, we train four ST-CoSense models at sampling rates δ in $\{0.2, 0.4, 0.6, 0.8\}$ and evaluate each model under all four rates, forming a 4×4 cross-evaluation matrix. We define the Generalization Ratio (GR) to quantify transferability:

$$\text{GR}(\delta_i, \delta_j) = \frac{\text{MAE}(\text{train} = \delta_i, \text{test} = \delta_i)}{\text{MAE}(\text{train} = \delta_i, \text{test} = \delta_j)}. \quad (16)$$

GR quantifies how well a model trained at sampling rate δ_i transfers to a different testing rate δ_j . A value close to 1.0 indicates that the rate mismatch causes little performance loss, while a value significantly below 1.0 indicates that the model is sensitive to the change in sampling rate. As shown in Fig. 8, GR values along the diagonal and its immediate neighbors remain close to 1.0 on both datasets, demonstrating that a single trained model generalizes well to neighboring sampling rates. As the gap between training and testing rates increases, GR values decline gradually but remain still in a reasonable range. In practice, this means that practitioners do not need to retrain the model for every possible budget scenario. A single model trained at a representative sampling rate can be deployed under moderate budget variations without significant performance loss, and for larger deviations, lightweight fine-tuning is sufficient to restore near-optimal accuracy.

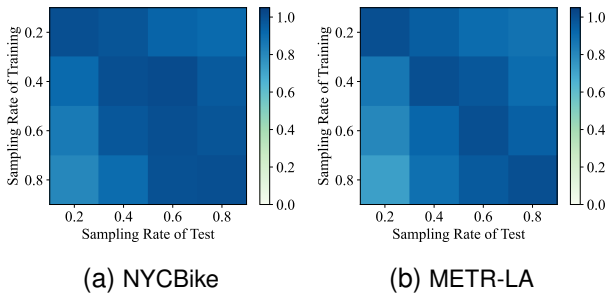


Fig. 8: Cross-rate generalization heatmaps on (a) NYCBike and (b) METR-LA.

The preceding analysis examines robustness when deployment conditions differ from training in terms of sampling rate.

A related practical concern is the availability of training data. The experiments in this work are conducted on benchmark datasets, where complete historical data is used as training supervision as specified in Eq. (14). To examine whether the framework can tolerate incomplete training data, we conduct an additional experiment in which the reconstruction loss is computed only at observed locations, removing the dependency on complete supervision. On NYCBike, this yields MAE of 1.98 and RMSE of 3.96, compared with 1.94 and 3.90 under full supervision, differences of 2.1% in MAE and 1.5% in RMSE. This minimal degradation is because the dynamic sampling policy ensures broad spatial coverage over the course of training, allowing every location to accumulate sufficient learning signals across batches. Additionally, for scenarios where higher training data quality is desired, practitioners can conduct short-term dense monitoring campaigns or leverage existing urban data archives to obtain near-complete observations prior to budget-constrained operation.

TABLE IV: The performance under different window lengths.

Datasets	$L = 6$	$L = 12$	$L = 24$	$L = 48$	$L = 96$
NYCBike	2.31	2.05	1.94	1.93	1.99
METR-LA	2.83	2.21	1.98	1.95	1.95

The time window size L is a critical hyperparameter that determines the amount of historical context available for both sampling decisions and spatio-temporal imputation. To evaluate its impact on model performance, we conduct experiments varying $L \in \{6, 12, 24, 48, 96\}$ on NYCBike and METR-LA datasets under 50% sampling rate.

Tab. IV reveals important patterns regarding the relationship between window size and performance. Very small windows (e.g. $L=6$) provide insufficient temporal context, resulting in substantially higher errors, MAE of 2.31 on NYCBike (19.1% degradation) and 2.83 on METR-LA (45.1% degradation) compared to optimal settings. The same L value corresponds to different absolute time spans: $L=6$ covers only 6 hours for NYCBike versus 30 minutes for METR-LA. While 6 hours may capture some temporal patterns, 30 minutes is too brief to capture traffic flow dynamics, explaining METR-LA's larger degradation.

Performance improves consistently as L increases from 6 to 48. At $L=24$, both datasets achieve near-optimal results, MAE of 1.94 for NYCBike and 1.98 for METR-LA. This setting covers 1 day for NYCBike, sufficient to capture daily mobility cycles including morning/evening peaks and day/night variations, and 2 hours for METR-LA, capturing short-term traffic dynamics and recurrent patterns within typical commute periods.

Further increasing to $L=48$ yields marginal improvements, achieving optimal performance with MAE of 1.93 for NYCBike and 1.95 for METR-LA. This extends coverage to 2 days for NYCBike to capture multi-day mobility patterns, and to 4 hours for METR-LA to better model within-day traffic dynamics. However, beyond $L=48$, the two datasets exhibit divergent behavior. NYCBike's performance degrades slightly at $L=96$ (MAE of 1.99), suggesting that extending the window beyond 2 days introduces noise or less relevant

historical information that dilutes the signal from recent, more predictive patterns. In contrast, METR-LA maintains stable performance at $L=96$ (MAE of 1.95), indicating that high-frequency traffic data benefits from longer temporal context. 8 hours of 5-minute observations provide rich information about traffic evolution without introducing the noise that affects lower-frequency data.

These findings suggest dataset-specific window size selection strategies. For hourly urban sensing data, $L=24$ provides an optimal balance between capturing relevant temporal patterns (daily cycles) and avoiding noise from distant history, while maintaining computational efficiency. For high-frequency applications, larger windows up to $L=48$ or even $L=96$ can be beneficial, as the increased temporal resolution provides denser information that outweighs potential noise. The key principle is to match the window size to the characteristic temporal scale of the phenomenon.

E. Spatio-Temporal Analysis (RQ4)

To visually demonstrate the spatial sensing capabilities of ST-CoSense, we compare its performance with GRIN on the NYCBike dataset at a 30% sampling rate (as shown in Fig. 6, this ratio lies in the low-ratio regime where performance differences among methods are discriminative, making 30% a suitable setting for visualization).

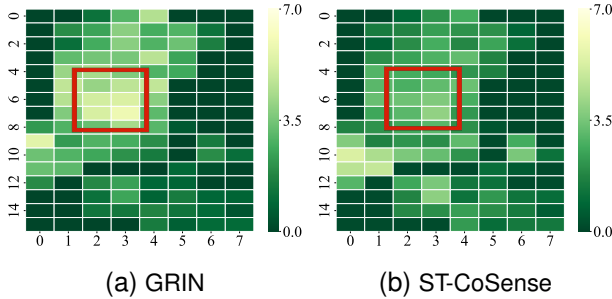


Fig. 9: Heatmap of spatial mean error at 30% sampling rate for NYCBike dataset: comparison between GRIN and ST-CoSense.

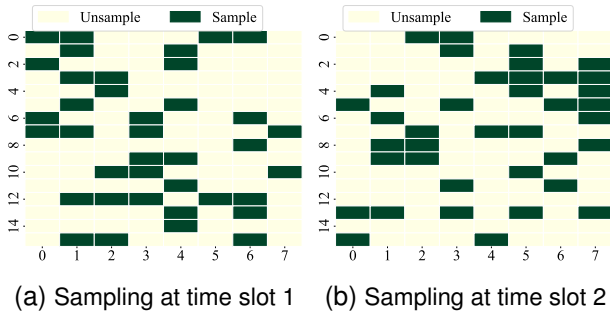


Fig. 10: Sample situation of adjacent moments.

As illustrated in Fig. 9, lighter colors represent larger imputation errors. Overall, ST-CoSense exhibits the darkest and most homogeneous error map, indicating better imputation performance at the 30% sampling rate. GRIN attains relatively

low errors owing to its graph-recurrent hybrid design, yet noticeable light-colored clusters persist in “hot spot” regions where data patterns are complex and rapidly changing. ST-CoSense further suppresses these localized errors. Its jointly optimized sampling strategy steers sensing resources toward such critical regions, while the coupled spatio-temporal imputer fully exploits cross-temporal correlations to reduce extreme deviations. This finding is consistent with the quantitative metrics reported in Tab. II and confirm ST-CoSense’s superior spatial awareness in highly dynamic urban areas.

Furthermore, we analyze the sampling distributions from two consecutive time slots and their impact on sensing performance, as depicted in Fig. 10. The sampling subareas in adjacent time slots exhibit complementary patterns, indicating that the strategy dynamically adapts to avoid redundant coverage. This temporal complementarity maximizes information acquisition efficiency across the spatial domain.

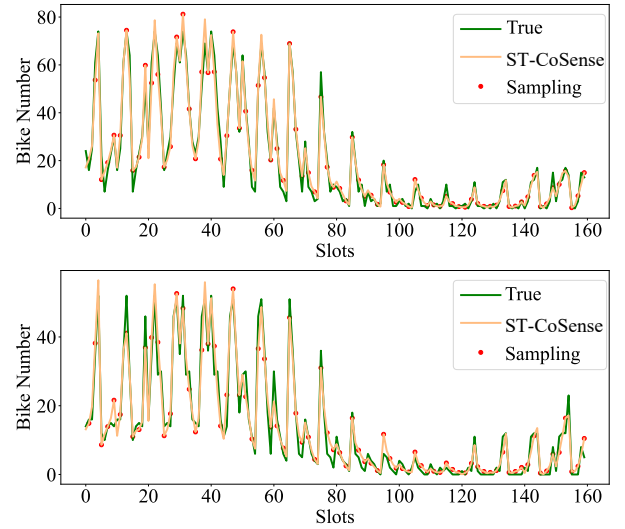


Fig. 11: Sensing performance for two subareas in the NYCBike Dataset at 30% Sampling Rate.

We also visualize the sensing performance of two subareas on the NYCBike dataset over time (the time dimensions of the two subareas are aligned). The results are shown in Fig. 11, where red dots indicate that this subarea is sampled during that time slot. Even with a sampling rate as low as 30%, ST-CoSense can still show a sensing trend similar to ground truth. The two subareas exhibit similar temporal trends within the same time window, confirming that inter-subarea correlations provide a reliable basis for spatial inference.

F. Training Efficiency (RQ5)

To evaluate training efficiency, we record the average training time per epoch on NYCBike for high-performing baselines. The results are presented in Tab. V. ST-CoSense consistently maintains competitive training efficiency.

This efficiency stems from two core design choices. First, by decoupling spatial and temporal modeling into separate stages, ST-CoSense avoids the costly joint spatio-temporal attention that operates over flattened sequences of length $|\mathcal{V}| \times L$.

Each module instead processes a much smaller input tensor independently, significantly reducing per-layer computation. Second, the ST encoder performs temporal compression before spatial reasoning begins, collapsing L time steps into a single compact representation per subarea. This early compression shrinks the tensor that downstream spatial attention consumes and lowers memory traffic during backpropagation. These two mechanisms together enable ST-CoSense to scale favorably while maintaining the modeling capacity required for accurate cooperative sensing.

TABLE V: Comparison of training time for each epoch.

Methods	ST-CoSense	GRIN	GSLI	CSDI	Imputeformer
Time(min)	0.51	0.68	0.31	2.60	1.12

VI. CONCLUSION

In this paper, we propose ST-CoSense, a unified framework for cost-efficient urban sensing. ST-CoSense integrates active sparse sampling and spatio-temporal imputation through end-to-end closed-loop optimization. Unlike prior decoupled approaches where sampling and imputation are optimized independently, ST-CoSense jointly trains both components so that the reconstruction loss guides sampling decisions via gradient-based feedback. The active sparse sampling module dynamically selects informative locations via a mask-guided encoder, while the spatio-temporal imputation module reconstructs complete urban fields through hierarchical spatial and temporal reasoning. This closed-loop design trains the sampler to select locations that minimize reconstruction error. Extensive experiments on four real-world urban datasets demonstrate that ST-CoSense achieves state-of-the-art performance, with up to 9.07% MAE and 10.53% RMSE improvements at a 50% sampling rate. On large-scale networks with thousands of nodes, the framework reduces RMSE by 4.95%, confirming its robustness in city-scale deployments. Future work will focus on relaxing the uniform sensing cost assumption, extending the framework to multi-attribute urban observations, and exploring cross-city transfer learning to further broaden its applicability.

VII. ACKNOWLEDGMENTS

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Yaxin Mei received the MS degree from Huaqiao University, Xiamen, China, in 2021. She is currently working toward the PhD degree with the School of Artificial Intelligence, Beijing Normal University, China. Her current research interests include urban computing, spatio-temporal data mining and mobile edge computing.



Yu Zheng (Fellow, IEEE) is currently the vice president with JD.COM, leading the JD Intelligent Cities Business Unit and JD Intelligent Cities Research. He is also the chief data scientist with JD Technology, passionate about using Big Data and AI technology to tackle urban challenges. He was a senior research manager with Microsoft Research. His research interests include Big Data analytics, spatio-temporal data mining, machine learning, and artificial intelligence.



Huiling Qin received the BE and PhD degrees from the School of Computer Science and Technology, Xidian University. Currently, she is a lecturer with the Institute of Artificial Intelligence and Future Networks, Beijing Normal University, China. Her research interests include urban computing, spatio-temporal data mining, and graph-based model.



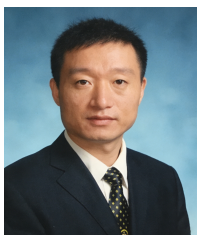
Tian Wang (Senior Member, IEEE) received the BSc and MSc degrees in computer science from Central South University, in 2004 and 2007, respectively, and the PhD degree in computer science from the City University of Hong Kong, in 2011. Currently, he is a professor with the Institute of Artificial Intelligence and Future Networks, Beijing Normal University. His research interests include Internet of Things, edge computing, and mobile computing. He has 30 patents and has published more than 300 papers in high-level journals and conferences. He was a co-recipient of the Best Paper Runner-up Award of IEEE/ACM IWQoS 2024. He has more than 18000 citations, according to Google Scholar. His H-index is 78.



Guangxue Zhang received the MS degree from Huaqiao University, Xiamen, China, in 2019. Currently, he is an engineer with the Institute of Artificial Intelligence and Future Networks, Beijing Normal University, China. His research interests include mobile edge computing, speech recognition.



Xiaolong Xu (Senior Member, IEEE) is currently a professor with the School of Computer and Software, Nanjing University of Information Science and Technology, Nanjing, China. His research interests include edge intelligence, recommendation systems, service computing, big data. He has 30 patents and has published more than 100 papers in high-level journals and conferences. He has more than 16000 citations, according to Google Scholar. His H-index is 70.



Zhanchuan Cai (Senior Member, IEEE) received the Ph.D. degree in computer software and theory from Sun Yat-sen University, Guangzhou, China, in 2007. He is currently a Professor with the School of Computer Science and Engineering, Macau University of Science and Technology, Taipa, Macau, China, where he is also with the State Key Laboratory of Lunar and Planetary Sciences. He has authored more than 200 papers in journals and refereed conferences. His research interests include intelligent information processing, image processing

and computer graphics, multimedia information security, and remote sensing data processing and analysis.